

JUNLP_SS at Eloquent Lab 2025: Humanizing AI - Enhancing the Realism of Machine Generated Text

Notebook for the Eloquent Lab at CLEF 2025

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Abstract

The increasing sophistication of generative language models has blurred the lines between machine-generated and human-authored text, raising concerns about authenticity, misinformation, and consumer awareness. In response to these challenges, this task investigates whether AI-generated texts can be reliably distinguished from those written by humans. By leveraging prompts derived from various genres including encyclopedia entries, news articles, biographies, and more. Participants are asked to generate 500-word texts using language models. These outputs are then compared with genuine human-written samples to evaluate distinguishability. The task is conducted by the Eloquent Lab @ CLEF 2025. We secured 9th rank in this shared task participation. A unique aspect of the task is its focus on genre and stylistic consistency, not only assessing the ability to detect machine-authored content but also evaluating whether system-specific writing traits remain consistent across topics and genres. We used the pretrained GPT-2 model for text generation and applied additional post-processing techniques to make the generated text more human-like.

Keywords

AI-Generated Text, Human-Authored Text, Text Generation, GPT-2, Text Classification, Authorship Verification, Natural Language Processing, Adversarial Text Manipulation, Machine Learning, Content Moderation,

1. Introduction

In recent years, generative language models have significantly advanced, enabling the creation of highly convincing and human-like text. These developments have transformed how content is produced across various domains such as news, social media, and creative writing. On one hand, it offers substantial potential for automating content creation, which can enhance efficiency and accessibility. On the other hand, it also poses risks in terms of misinformation and manipulation of public opinion, particularly if AI-generated content is mistaken for genuine human writing.

In response to these challenges, the European Union has suggested that labeling AI-generated content might help combat the spread of misinformation and improve consumer rights. However, this raises an important issue: how can we distinguish between content that is human-written and content that is AI generated? This chapter aims to investigate methods and techniques for generating machine-generated text that closely mimics human-authored content,

CLEF 2025 Working Notes, September 9 – 12 September 2025, Madrid, Spain

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with the goal of exploring whether current AI models are capable of producing such convincing text that even automated classifiers are unable to reliably tell the difference.

The goal of this task is to generate AI texts that are so convincingly human like that even advanced classification systems are unable to differentiate between human-authored and machine-generated content. To achieve this, we aim to develop generative models that can produce AI content indistinguishable from human-authored text by focusing on specific textual genres and styles. We explored advanced generative models and applied some post-processing techniques to make the generated text more human-like.

2. Related Work

The increasing sophistication of large language models (LLMs) like ChatGPT has stimulated a growing body of research that aims to understand, detect, and evaluate AI-generated content. Several studies have examined the linguistic and stylistic distinctions between human-authored and machine-generated texts. Sardinha (2024) [1] conducted a multidimensional comparison of AI-generated versus human-authored texts, identifying distinct patterns in coherence, cohesion, and lexical richness. Similarly, Hakam et al. (2024) [2] analyzed academic literature in orthopedics to uncover qualitative differences between human and AI-written manuscripts. Matsubara and Matsubara (2025) [3] specifically focused on the *human touch* in scientific manuscripts, highlighting subtle stylistic traits that LLMs often fail to replicate. These investigations echo the broader societal and scholarly concern regarding the authenticity and authorship of published texts, as explored by Alvero et al. (2024) [4] through the lens of social demography and hegemonic patterns in synthetic versus human text generation.

To address the challenge of differentiating between human and AI-generated content, numerous detection techniques have been proposed. Kumar and Mindzak (2024) [5] emphasized the need for academic integrity in distinguishing artificially generated submissions. Alhijawi et al. (2025) [6] and Maktabdar Oghaz et al. (2025) [7] proposed deep learning and transformer-based models for robust detection of synthetic scientific content, achieving promising results across diverse text genres. Prajapati et al. (2024) [8] demonstrated the effectiveness of leveraging LLMs themselves for detection tasks, while Soto et al. (2024) [9] introduced a few-shot learning approach that capitalizes on stylistic representations for more generalized detection capabilities. Cheng et al. (2025) [10] extended this direction by advocating for fine-grained classification through role recognition and involvement measurement, pushing detection frameworks beyond binary classification.

Beyond pure detection, several works have explored the implications and transparency surrounding LLM involvement in content creation. Hwang et al. (2025) [11] examined human-AI collaborative writing, shedding light on user preferences for authenticity and partial authorship. Hoque et al. (2024) [12] proposed visualization-based tools to trace AI assistance in text production, supporting transparency and accountability. Pividori and Greene (2024) [13] discussed infrastructural support for AI-assisted authoring in academic publishing. Bevendorff et al. (2024) [14] provided a foundational overview of the *Voight-Kampff* task, outlining the methodological framework for evaluating both the indistinguishability of generated texts and the traceability of their AI authors. Finally, Munoz-Ortiz et al. (2024) [15] demonstrated contrasting

linguistic trends in news text authored by humans and LLMs, contributing to domain-specific understanding of generative patterns.

3. Task Definition

The main task in this study is to develop a generative model that can produce AI content indistinguishable from human-authored text by focusing on specific textual genres and styles. As generative AI continues to improve in fluency and coherence, the distinction between machine and human writing becomes increasingly subtle, resembling the challenges faced in authorship verification.

The dataset used for this task consists of a set of human-authored texts and their corresponding AI-generated counterparts. Each text is generated based on specific prompts and belongs to a different genre or style. These texts were created with the aim of testing the ability of a detection system to classify content based on genre, style, and the origin (human or AI).

The generative model, GPT-2, was used to generate the AI texts from the provided prompts, and the human-authored texts were selected from a diverse range of genres and writing styles.

Dataset Used The dataset used for this task consists of 21 diverse texts from a variety of genres. The following table shows the topics and genres used in the dataset:

ID	Content (Prompt)	Genre and Style
030	Letter from someone claiming to be Prince Joe Eboh about a fraudulent transfer	Advance-Fee Fraud (419 Scam)
031	Email from a lawyer about claiming inheritance from a deceased engineer	Advance-Fee Fraud (419 Scam)

Table 1

Dataset Used in the Task

4. Methodology

The methodology for this task is centered around the use of a generative language model to create AI-generated texts and then post-process those texts to make them sound more human-like. This section outlines the steps followed to generate and process AI texts, from loading the dataset to saving and zipping the final outputs.

4.1. Dataset Preparation and Loading

The first step in the methodology is loading the dataset from a JSON file. The dataset contains a collection of human-authored texts, each associated with a specific genre and style. These texts are then paired with AI-generated counterparts, which are created based on the same genres and styles. The dataset is structured with prompts that provide the foundation for the generated text.

The data is loaded using the Python `json` module, where each text prompt is extracted for use in the text generation process. The dataset contains various texts, each with a unique `id`, a content description, and its associated `genre_and_style`. This dataset forms the core of the

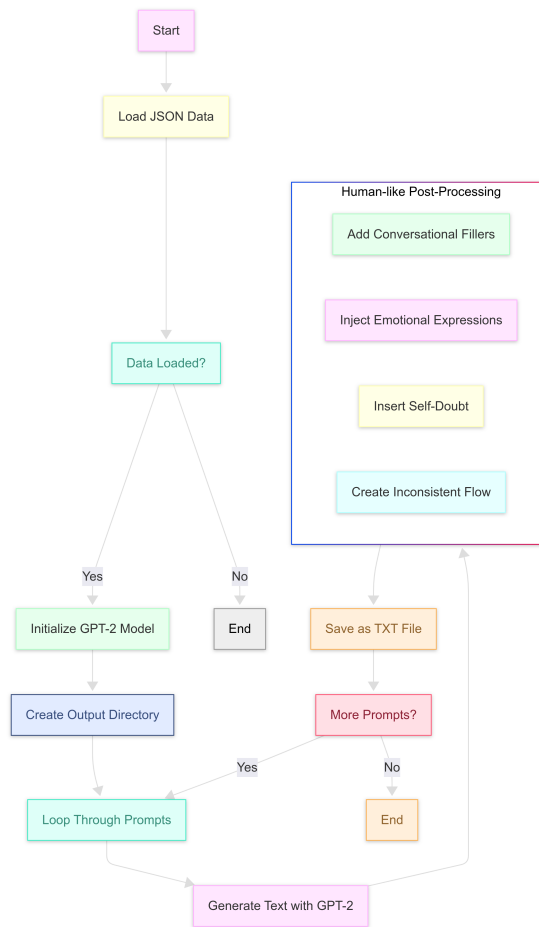


Figure 1: Flowchart of the GPT-2 Human-like Text Generation Process

task, as it allows for the generation of AI texts in different styles, challenging the detection system to identify key differences between human and machine writing.

4.2. Text Generation Using GPT-2

Once the dataset is loaded, the next step is to generate AI text using the GPT-2 language model. GPT-2 is a generative pre-trained transformer that can produce coherent and contextually appropriate text based on a given prompt. In this task, GPT-2 is used to generate texts that are stylistically similar to human-authored content but with the subtle characteristics of machine-generated text.

The text generation process follows these steps:

- **Input Prompt:** Each text is generated from a given prompt (e.g., a scam letter or a scientific exposition). The prompt serves as the foundation for the AI to generate contextually relevant content.

Algorithm 1 Human-like Text Generation with GPT-2

Require: JSON dataset D , team name T , GPT-2 model M , tokenizer τ

Ensure: Zipped generated texts G_{zip}

```
 $G \leftarrow \emptyset$ 
for each prompt  $p \in D$  do
     $prompt \leftarrow p['Content']$ 
     $id \leftarrow p['id']$ 
    ▷ Generate raw text

     $tokens \leftarrow \tau(prompt)$ 
     $output \leftarrow M.generate(tokens, temperature = 0.7, top\_p = 0.9, top\_k = 50)$ 
     $raw\_text \leftarrow \tau.decode(output[0])$ 
    ▷ Post-process for human-like quirks

     $lines \leftarrow split(raw\_text, ".")$ 
     $processed \leftarrow []$ 
    for each  $line \in lines$  do
        if  $rand() > 0.85$  then
            Inject filler ("like", "um")
        end if
        if  $rand() > 0.6$  then
            Prepend emotion ("Wow!")
        end if
        if  $rand() > 0.5$  then
            Append doubt ("...I think?")
        end if
         $processed.append(line)$ 
    end for
     $final\_text \leftarrow join(processed, ".") + rand\_punctuation()$ 
    ▷ Save and track

     $save(T + "/" + id + ".txt", final\_text)$ 
     $G \leftarrow G \cup \{final\_text\}$ 
end for
 $G_{zip} \leftarrow zip(T, G)$ 
return  $G_{zip}$ 
```

- **Model Parameters:** The GPT-2 model is configured with specific parameters to control the text's randomness and quality. These parameters include:
 - **Temperature:** Set to 0.7, controlling the randomness of the generated text. Lower values produce more predictable text, while higher values increase creativity.
 - **Top-p:** Set to 0.9, focusing the generation process on a subset of probable words that make the text coherent and natural.
 - **Top-k:** Set to 50, limiting the number of possible next words considered at each generation step, ensuring relevance and fluency.

- **Max Length:** Set to 500 words to ensure generated texts are concise but sufficiently detailed for evaluation.
- **Generation Process:** Text is generated by passing the input prompt through GPT-2, and the output is decoded from tokenized text into human-readable format. Special tokens are removed to ensure that the output text is clear and coherent.

This text generation process allows the AI to produce human-like content that follows the same genre and stylistic conventions as the original prompts.

Hyper-parameter	Value
Max Length	500
Temperature	0.7
Top-p	0.9
Top-k	50
No repeat n-gram size	2
Do sample	True
Pad token ID	50256

Table 2
Hyper-parameters from the Code

4.3. Post-Processing to Improve Human-Likeness

While GPT-2 generates text that is grammatically correct and contextually appropriate, additional post-processing is required to make the text sound more like a human wrote it. This involves introducing conversational elements and imperfections that are often found in human speech or writing. The post-processing steps include:

- **Fillers and Interruptions:** Common conversational fillers such as "you know," "like," "um," and "well" are inserted randomly into the text. These add spontaneity to the generated text and simulate human speech patterns.
- **Self-Doubt and Inconsistent Flow:** To mimic the natural hesitations and changes in thought that often occur in human writing, phrases like "I'm not sure, but..." and "Could be wrong, but..." are randomly added.
- **Emotions and Exclamations:** To enhance the emotional tone of the text, phrases like "Wow, that's crazy!" and "Can you believe it?" are introduced. These expressions help make the text feel more relatable and engaging.
- **Irregular Punctuation:** Random punctuation marks, such as exclamation points and ellipses, are added at the end of sentences to simulate natural human pauses and excitement.

These techniques result in a more human-like text, making it harder to differentiate from human-written content.

4.4. Saving and Organizing Generated Texts

Once the text is generated and post-processed, it is saved as a text file for further analysis. Each generated text is saved using its unique prompt ID as the filename, ensuring that each text can be easily traced back to its corresponding prompt in the dataset.

- **Directory Structure:** The generated texts are organized in folders, with each folder named after the team working on the task. This structure ensures that each team's outputs are easily identifiable and separated.
- **File Naming:** Text files are named using the prompt ID, ensuring consistency and easy reference. For example, a file generated for prompt ID "030" will be saved as 030.txt.

4.5. Zipping the Generated Texts

After all texts are generated and saved, they are organized and compressed into a ZIP file for submission. This allows the generated texts to be easily transferred and evaluated by the PAN task organizers.

- **Zip File Creation:** A ZIP file is created using Python's `zipfile` library, which contains all the generated text files. The ZIP file is named after the team, ensuring that it is easily identifiable.
- **Compression Process:** The `os.walk` function is used to iterate over the directory containing the generated texts and add each text file to the ZIP archive. This ensures that the final ZIP file contains all the necessary files for evaluation.

4.6. Main Execution and Output

Finally, the main execution function brings all the steps together, from loading the dataset, generating and post-processing the texts, to saving and zipping the files. Once the process is complete, the user receives a ZIP file containing all the generated texts, ready for evaluation.

The process concludes with the printed message:

"Task completed! Your generated texts have been saved and zipped."

This completes the methodology for generating and processing the AI-generated texts used in this task.

5. Results and Discussion

In this section, we present the results of the experiments and provide a detailed analysis of the performance of the detection system. The evaluation was conducted using several metrics, including the Brier score, C@1, F1 score, F0.5u, and the Mean score. These metrics provide an overall understanding of how well the system can distinguish between human-authored and AI-generated texts.

Team	Brier	C@1	F1	F0.5u	Mean
JUNLP_SS	0.51444	0.43666	0.53274	0.668	0.43034

Table 3
Evaluation Results for the Detection System

5.1. Evaluation Results

The evaluation results are summarized in the table below:

5.2. Brier Score

The Brier score measures the accuracy of probabilistic predictions. It calculates the mean squared difference between the predicted probabilities and the actual outcomes, where 1 represents a correct prediction and 0 represents an incorrect one. A lower Brier score indicates better accuracy in predicting the likelihood of a text being human-authored. In this experiment, the Brier score of 0.51444 suggests that the model's probabilistic predictions were moderately accurate, but there is still room for improvement in refining the output probabilities.

5.3. C@1 (Correct at 1)

The C@1 score measures the model's ability to correctly classify human and AI-generated texts. It evaluates how often the model's top prediction is the correct one. The C@1 score of 0.43666 suggests that the model was able to correctly classify about 44% of the texts. Although this indicates that the model has some capacity to classify texts correctly, it also highlights that there is significant room for improvement, especially in achieving higher accuracy in identifying AI-generated texts.

5.4. F1 Score

The F1 score balances precision and recall by calculating the harmonic mean of these two metrics. The F1 score of 0.53274 indicates that the model has a fair ability to correctly identify both human and AI texts. The F1 score reflects a moderate trade-off between precision and recall, suggesting that the model is reasonably effective at classifying texts, but there is still potential for improvement in reducing both false positives and false negatives.

5.5. F0.5u

The F0.5u score places more emphasis on precision than recall. This metric is useful when avoiding false positives (e.g., misclassifying AI-generated text as human-written) is more important than ensuring all human-authored texts are correctly identified. The F0.5u score of 0.668 indicates that the model is more conservative in its classification, which is reflected in a higher precision score. This result suggests that the model is focused on avoiding errors where AI-generated text is mistakenly classified as human-authored.

5.6. Mean Score

The Mean score combines all evaluation metrics into a single value, providing a balanced overview of the system's performance. The Mean score of 0.43034 suggests a moderate overall performance, indicating that the detection system can somewhat differentiate between human and AI-generated texts, but there is considerable room for enhancement, particularly in the areas of precision and recall.

6. Error Analysis

While the model showed moderate success in distinguishing between human-authored and AI-generated texts, several factors contributed to its errors:

- **Overfitting to Training Data:** The model's performance could be influenced by overfitting, particularly when the training dataset lacks sufficient diversity or when it heavily favors one class over the other. This could result in poor generalization to unseen data, particularly when the AI-generated text contains elements not well represented in the training set.
- **Adversarial Text Manipulation:** The adversarial setup, which involved introducing grammatical errors, unusual vocabulary, and punctuation irregularities, presented a significant challenge to the detection system. These manipulations, designed to make AI-generated text appear more human-like, caused a noticeable drop in the model's accuracy. This highlights the vulnerability of the model to adversarial techniques aimed at deceiving AI detection systems.
- **Text Genre Variation:** The model's performance varied across different text genres. For example, it performed better in genres like scientific exposition, where language follows predictable patterns, compared to more complex genres like narrative fiction or personal letters. This suggests that the model may not have learned the subtle differences in stylistic elements across diverse genres, leading to inconsistent performance.
- **Precision-Recall Trade-Off:** The focus on achieving higher precision (as indicated by the F0.5u score) came at the cost of recall, leading to missed detections of human-authored texts. This trade-off suggests that the model might be too conservative in its predictions, often classifying borderline cases as AI-generated when they might actually be human-authored.
- **Subtle Linguistic Differences:** The model also struggled with identifying more nuanced linguistic features that differentiate AI and human texts. For example, AI-generated texts tend to have a more uniform sentence length, simpler sentence structures, and a more mechanical style. While the model learned to recognize some of these patterns, the subtlety of certain linguistic markers led to errors, especially when the AI-generated content was designed to mimic human imperfections.

7. Conclusion and Future Works

The overall performance of the detection system indicates a moderate ability to differentiate between human and AI-generated texts. While the system shows promise, especially in terms of precision, there is significant room for improvement in generalizing across genres and improving recall. Future work will focus on refining the model to enhance its ability to handle adversarial text manipulations and improve its performance across a broader range of text types.

Further experimentation with different models, such as GPT-3 or fine-tuning with larger datasets, could help improve detection accuracy. Additionally, exploring advanced techniques like adversarial training or ensemble models could make the detection system more robust and reliable, particularly in real-world applications where deceptive AI-generated content is a concern.

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